

Git support. A Git repository can be created as a master branch, where other programmers can check out their own branch as a local copy, work on it and then push back their code. The code can then be merged with the master branch by the project leads who are maintaining the code repository.

tmate

tmate (<https://tmate.io>) is used for instant terminal sharing. It is a fork of *tmux*, and can be used to share a *ssh* session with other collaborating workers, simultaneously, on the same shell, by just sharing an *ssh* link that comes at the bottom of the terminal when starting tmate. Once installed, you can launch it with *tmate*. You should see something like what's shown below:

```
ssh PMhmes4XeKQyBR2Jtvt6BJw@nyc1.
tmate.io
```

This allows others to join your terminal session. All users see the same terminal content at all times. This

is useful for pair programming, when two people share the same screen but have different keyboards.

OpenVPN

This solution enables a user to get access to an Intranet remotely. This and many other solutions can be installed on-premise and can be managed centrally. OpenVPN allows peers to authenticate each other using pre-shared secret keys, certificates or user names/ passwords. When used in a multi-client-server configuration, it allows the server to release an authentication certificate

for every client, using signatures and certificate authority. It uses the OpenSSL encryption library extensively, as well as the TLS protocol, and has many security and control features.

If you install FreedomBox, many solutions like OpenVPN, Matrix, etc, can be installed easily through FreedomBox's Plinth interface.

So this is the right time to try these ethical, privacy-friendly software services in government firms and offices that care about not only their own privacy but also the privacy of their clients and other stakeholders. **END** 

References

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One of the greatest challenges is controlling or removing quantum decoherence. This usually means isolating the system from its environment as interactions with the external world cause the system to decohere. However, other sources of decoherence also exist. Examples include the quantum gates, and the lattice vibrations and background thermonuclear spin of the physical system used to implement the qubits. Decoherence is irreversible.

Splitting the photon into three beams through the use of an intricate contraption consisting of the calibrated setup of lasers, beam splitters and barium borate crystals, the researchers created qutrits. They also said that it could be possible to use ququarts in the future. The researchers were

able to prove that qutrit teleportation is possible. Despite rapid and impressive experimental progress, most researchers believe that “fault-tolerant quantum computing [is] still a rather distant dream.”

No scalable quantum computing hardware had been demonstrated till September 2019. Nevertheless, there is an increasing amount of investment in quantum computing by governments, established companies, and startups.

Current research focuses on building and using near-term intermediate-scale devices and demonstrating quantum supremacy alongside the long-term goal of building and using a powerful and error-free quantum computer.

Quantum based computers will arrive in the market sooner than we think. What is most exciting is not what we can do with a quantum computer today, but the undiscovered truths it will reveal tomorrow! **END** 

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The author has previously worked with the Defence Research and Development Organization (DRDO) as a scientist. He has been awarded medal and citation by former Prime Minister Indira Gandhi for his contribution as chief design engineer of Arjun main battle tank, which has been accepted by Defence Forces and has gone into production since then.

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